

A purple square with the word "HTML" in white, bold, sans-serif capital letters.

HTML

HTML Complete Study Notes

Structured reference for learning and revision

Clean reference • Study notes • Interview preparation

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Goal: Learn HTML, understand what each tag/attribute does, and have a clean reference to return to while studying or preparing for interviews.

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1. What is HTML?

HTML = HyperText Markup Language.

HTML is the standard markup language used to create web pages.

It describes the structure of a web page and consists of a series of elements.

Why learn HTML?

From the supplied session material:

- 1 Understand how the frontend works.
- 1 Build and test APIs easily.
- 1 Debug issues between frontend and backend.

2. Tags and Elements

What is an HTML Tag?

A tag is the markup used to tell the browser what an element is.

Most tags come in pairs:

```
<p>This is a paragraph.</p>
```

Here:

- 1 `<p>` = opening tag
- 1 `</p>` = closing tag
- 1 `This is a paragraph.` = content

What is an HTML Element?

An element can contain:

```
Start tag + Content + End tag
```

Example:

```
<p>Hello</p>
```

Some elements have no content and therefore have no closing tag.

Examples:

```
<br>  
<hr>  
<img>  
<input>
```

These are commonly called empty elements.

Nested Elements

A nested element is an element placed inside another element.

```
<a href="https://github.com">  
    
</a>
```

Here:

- 1 `<a>` is the parent.
- 1 `<img>` is the child.
- 1 `<img>` is nested inside `<a>`.

Do Not Duplicate an Attribute

Do not repeat the same attribute inside one tag.

```
<!-- Wrong -->

```

The supplied revision material states that the browser uses the first `src` and ignores the repeated one.

3. Basic HTML Structure

Every HTML page follows this basic structure:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Page Title</title>
  </head>

  <body>
    <!-- Visible content -->
  </body>
</html>
```

Part	Purpose
<code><!DOCTYPE html></code>	Declares an HTML5 document
<code><html></code>	Root element
<code><head></code>	Metadata
<code><body></code>	Visible page content

Important

`<body>` contains visible content such as:

- 1 Headings
- 1 Paragraphs
- 1 Images
- 1 Hyperlinks
- 1 Tables
- 1 Lists

4. Head Tag

The `<head>` element is a container for metadata.

It can contain:

- 1 `<title>`
- 1 `<meta>`
- 1 Other information used by the browser/document

`<title>`

Defines the title of the document.

```
<title>My Website</title>
```

<meta>

The supplied material lists these uses:

- 1 Character set
- 1 Page description
- 1 Keywords
- 1 Author
- 1 Viewport settings

Example:

```
<meta charset="UTF-8">

<meta
  name="description"
  content="HTML study notes"
>

<meta
  name="keywords"
  content="HTML, Web Development"
>

<meta
  name="author"
  content="Student"
>

<meta
  name="viewport"
  content="width=device-width, initial-scale=1.0"
>
```

5. Text and Content

Headings — <h1> to <h6>

Headings are titles/subtitles displayed on a webpage.

```
<h1>Main Title</h1>
<h2>Section Title</h2>
<h3>Sub-section</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>
```

<h1> is the highest level and <h6> is the lowest level.

SEO Note

The supplied revision sheet recommends using one <h1> as the main topic of the page.

Paragraph — <p>

Used to define a paragraph.

```
<p>This is a paragraph.</p>
```

A paragraph starts on a new line and is usually a block of text.

The browser automatically removes extra spaces and lines when the page is displayed.

Lorem Ipsum

Lorem Ipsum is placeholder text used while building a layout when the real content is not ready.

In VS Code:

```
lorem
```

then press Tab.

You can also request a number of words:

```
lorem10  
lorem50  
lorem100
```

Line Break —

Creates a new line.

```
<p>  
  Line one<br>  
  Line two  
</p>
```


 is an empty element.

Horizontal Rule — <hr>

Creates a horizontal dividing line.

```
<hr>
```

<hr> is an empty element.

Text Formatting

Tag	Purpose	Example
<code></code>	Bold	<code>Important</code>
<code><i></code>	Italic	<code><i>Emphasized</i></code>
<code><u></code>	Underline	<code><u>Underlined</u></code>
<code><mark></code>	Highlight	<code><mark>Important</mark></code>
<code></code>	Strikethrough	<code>\$100</code>
<code><sup></code>	Superscript	<code>x<sup>2</sup></code>
<code><sub></code>	Subscript	<code>H<sub>2</sub>O</code>

Examples:

```
<p>Water = H<sub>2</sub>O</p>

<p>Area = r<sup>2</sup></p>

<p>
  Old price:
  <del>$100</del>
  New price:
  <mark>$70</mark>
</p>
```

`<font>` Tag

The old `<font>` tag was used to change text color, size, and face:

```
<font color="red" size="4">
  This text is red
</font>
```

The supplied material notes that `<font>` is old-style and should be avoided in modern HTML in favor of CSS.

6. Links

Anchor — `<a>`

Creates a hyperlink.

```
<a href="https://www.google.com">
  Visit Google
</a>
```

href

Specifies the destination.

```
<a href="https://github.com">
  GitHub
</a>
```

target

Specifies where the linked document opens.

Value	Behavior
<code>_self</code>	Same tab; default
<code>_blank</code>	New tab

```
<a
  href="https://github.com"
  target="_blank"
>
  Open GitHub
</a>
```

Clickable Image

Put `<img>` inside `<a>`:

```
<a
  href="https://github.com/Eyadeltaher/Cybersecurity-notes"
  target="_blank"
>
  
</a>
```

The `<a>` is the parent and `<img>` is the child.

7. Images

`<img>`

Embeds an image.

```

```

Important Attributes

Attribute	Purpose
<code>src</code>	Image path
<code>alt</code>	Alternative text
<code>width</code>	Display width

Why is `alt` important?

The supplied revision sheet gives two main reasons:

- 1 Helps search engines understand the image.

-
- 2 Helps screen readers describe the image.

Good:

```
alt="red sports car"
```

Not useful:

```
alt="image1"
```

8. Comments

HTML comments:

```
<!-- This is a comment -->
```

Example:

```
<!-- Navigation section -->
<nav>
  <a href="#">Home</a>
</nav>

<!-- TODO: add dropdown menu here -->
```

Comments are not displayed as normal page content.

9. Layout and Containers

<div>

Defines a division or section.

It is commonly used as a block-level container for grouping elements.

```
<div style="background-color: lightblue;">
  <h2>Section Title</h2>
  <p>Content goes here.</p>
</div>
```


An inline container.

Useful for styling or targeting a small part of text.

```
<p>
  My favorite color is
  <span style="color: red;">red</span>.
</p>
```

Easy way to remember

```
<div> → block / box
<span> → inline / small part
```

10. Semantic HTML

Semantic elements clearly describe their meaning to the browser and developer.

Examples:

```
<header>
<main>
<footer>
<nav>
<section>
```

<header>

Container for introductory content.

<main>

Contains the main content of the document.

<footer>

Defines a footer for a document or section.

<nav>

Defines navigation links.

<section>

Defines a section in a document.

11. Media

Image

```

```

Video — <video>

Used to show a video.

```
<video
  src="movie.mp4"
  width="600"
  controls
  muted
  autoplay
  loop
>
  Your browser does not support video.
</video>
```

Video attributes

Attribute	Purpose
<code>controls</code>	Shows playback controls
<code>muted</code>	Starts without sound
<code>autoplay</code>	Plays automatically
<code>loop</code>	Repeats

The supplied notes mention that browsers often block autoplay with sound, so `muted + autoplay` is commonly used.

Audio — `<audio>`

Used to play audio.

```
<audio
  src="song.mp3"
  controls
  autoplay
  loop
  muted
></audio>
```

Iframe — `<iframe>`

Embeds another webpage or external content.

Examples include YouTube and Google Maps.

```
<iframe
  src="https://www.youtube.com/embed/dQw4w9WgXcQ"
  width="560"
  height="315"
>
</iframe>
```

Easy way to remember

An iframe is like a window inside your webpage that displays external content.

12. Lists

HTML has three main list types:

- 1 Ordered list
- 2 Unordered list
- 3 Description list

Ordered List — `<ol>`

Shows items in an order.

```
<ol>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

type and start

```
<ol type="A" start="3">
  <li>Step C</li>
  <li>Step D</li>
</ol>
```

type examples:

- 1 1 numbers
- 1 A uppercase
- 1 a lowercase
- 1 I Roman numerals

start controls the starting value.

Unordered List —

Shows bullet points.

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

Bullet types

Type	Style
disc	
circle	
square	

Description List — <dl>

Used for terms and their descriptions.

```
<dl>
  <dt>HTML</dt>
  <dd>The structure language of the web</dd>

  <dt>CSS</dt>
  <dd>The styling language for web pages</dd>
</dl>
```

Tag	Meaning
<dl>	Description list
<dt>	Term

Tag	Meaning
<dd>	Description

13. Tables

A table contains cells arranged in rows and columns.

```
<table border="1">
  <thead>
    <tr>
      <th>Name</th>
      <th>Age</th>
    </tr>
  </thead>

  <tbody>
    <tr>
      <td>Ahmed</td>
      <td>25</td>
    </tr>
  </tbody>

  <tfoot>
    <tr>
      <td colspan="2">End of Table</td>
    </tr>
  </tfoot>
</table>
```

Table Tags

Tag	Purpose
<table>	Table wrapper
<thead>	Header section
<tbody>	Main content
<tfoot>	Footer section
<tr>	Table row
<th>	Header cell
<td>	Data cell

Important Attributes

border

```
<table border="1">
```

Adds a border.

colspan

Merges columns horizontally.

```
<td colspan="2">Full Name</td>
```

rowspan

Merges rows vertically.

```
<td rowspan="2">Ahmed</td>
```

Memory Trick

```
colspan → columns → horizontal  
rowspan → rows → vertical
```

14. Forms

A form collects user input.

```
<form  
  action="/submit"  
  method="POST"  
>  
  <!-- inputs -->  
  <button type="submit">Send</button>  
</form>
```

A form can contain:

- 1 <input>
- 1 <label>
- 1 <select>
- 1 <textarea>
- 1 <datalist>
- 1 <button>

15. Input Types

Text

```
<input  
  type="text"  
  placeholder="Enter your name"  
  required  
>
```

Single-line text.

Password

```
<input type="password">
```

Password field.

Email

```
<input
  type="email"
  placeholder="example@mail.com"
>
```

Email input.

File

```
<input type="file">
```

File selection/upload input.

Number

```
<input
  type="number"
  min="1"
  max="100"
>
```

Numeric input.

Range

```
<input type="range">
```

Used when the exact numeric value is not important.

Search

```
<input type="search">
```

Search field.

Telephone

```
<input
  type="tel"
  placeholder="+20 100 000 0000"
>
```

Telephone number.

Why `tel` instead of `number`?

The supplied revision sheet recommends `tel` for phone numbers because phone numbers can contain:

- 1 +
- 1 spaces
- 1 leading zeros

Phone numbers should be treated as contact information rather than mathematical numbers.

Date

```
<input type="date">
```

Time

```
<input type="time">
```

Week

```
<input type="week">
```

Color

```
<input type="color">
```

Radio

Allows one selection from a group.

```
<input
  type="radio"
  name="gender"
  value="male"
>
Male

<input
  type="radio"
  name="gender"
  value="female"
>
Female
```

Checkbox

Allows zero or more selections.

```
<input
  type="checkbox"
  name="hobby"
  value="reading"
>
Reading

<input
  type="checkbox"
  name="hobby"
  value="coding"
>
Coding
```

Submit

```
<input type="submit" value="Send Form">
```

Submits form data.

Reset

```
<input type="reset">
```

Resets form values to their defaults.

16. Form/Input Attributes

placeholder

Shows a short hint inside the field.

```
<input  
  type="text"  
  placeholder="Your full name"  
>
```

The hint disappears when the user types.

required

Makes the field required.

```
<input  
  type="email"  
  required  
>
```

The form cannot be submitted while the required field is empty.

readonly

Makes the input read-only.

```
<input  
  type="text"  
  value="Read only"  
  readonly  
>
```

disabled

Disables the input.

```
<input  
  type="text"  
  value="Disabled"  
  disabled  
>
```

17. Radio and Checkbox Grouping

Radio Buttons

The important rule:

Radio buttons that belong to the same group should have the same name and different value values.

Correct:

```
<input
  type="radio"
  name="color"
  value="red"
>
Red

<input
  type="radio"
  name="color"
  value="blue"
>
Blue

<input
  type="radio"
  name="color"
  value="green"
>
Green
```

Because they have the same **name**, selecting one deselects the others.

Wrong

```
<input
  type="radio"
  name="color1"
  value="red"
>
Red

<input
  type="radio"
  name="color2"
  value="blue"
>
Blue
```

Different names make them independent.

Checkbox

Checkboxes can have the same **name** when they represent related options.

```

<input
  type="checkbox"
  name="hobby"
  value="reading"
>
Reading

<input
  type="checkbox"
  name="hobby"
  value="coding"
>
Coding

```

18. Common Attributes

Attribute	Purpose
src	Resource/image path
alt	Alternative image text
href	Link destination
target	Link opening location
width	Display width
style	Inline CSS
id	Identifier
class	Reusable class
name	Field name / radio grouping
value	Input value
type	Input/list type
placeholder	Input hint
required	Required field
readonly	Read-only field
disabled	Disabled field
action	Form submission destination
method	Form submission method
min	Minimum numeric value
max	Maximum numeric value
rows	Textarea rows
cols	Textarea width
start	Ordered-list starting value
colspan	Merge table columns
rowspan	Merge table rows
controls	Media controls
muted	No sound
autoplay	Automatic playback
loop	Repeat playback

19. Other Form Elements

<label>

Defines a label for a form element.

```
<label for="email">
  Email:
</label>

<input
  type="email"
  id="email"
>
```

<select>

Creates a dropdown list.

```
<select name="country">
  <option value="egypt">Egypt</option>
  <option value="usa">USA</option>
</select>
```

<option>

Defines one selectable option inside <select>.

<textarea>

Creates a multiline input.

```
<textarea
  rows="5"
  cols="40"
  placeholder="Write your message..."
></textarea>
```

<datalist>

Provides predefined options for an input.

```
<input list="browsers">

<datalist id="browsers">
  <option value="Chrome">
  <option value="Firefox">
  <option value="Edge">
</datalist>
```

<button>

Creates a clickable button.

```
<button type="submit">
  Send
</button>
```

20. Form Attributes

action

Defines where/what the form submission should be sent to.

```
<form action="/submit">
```

method

Specifies the HTTP method used when submitting form data.

The supplied session material lists:

1 GET

1 POST

Example:

```
<form
  action="/submit"
  method="POST"
>
```

21. id vs class

id

An **id** is intended to uniquely identify an element.

```
<div id="main-section">
  Main content
</div>
```

class

A class can be used by multiple elements.

```
<div class="card">Card 1</div>
<div class="card">Card 2</div>
```

Easy rule

```
id    → unique identifier
class → reusable group
```

22. Complete Example

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta
    name="description"
    content="HTML study page"
  >

  <meta
    name="keywords"
    content="HTML, Web Development"
  >

  <meta
    name="author"
    content="Student"
  >

  <meta
    name="viewport"
    content="width=device-width, initial-scale=1.0"
  >

  <title>HTML Study Page</title>

</head>

<body>

  <header>

    <h1>HTML Study Page</h1>

    <nav>
      <a href="#content">Content</a>
      <a href="#media">Media</a>
      <a href="#form">Form</a>
    </nav>
```

```
</header>

<main>

  <section id="content">

    <h2>Text</h2>

    <p>
      This is a paragraph.
    </p>

    <p>
      <b>Bold</b>
      <i>Italic</i>
      <u>Underline</u>
      <mark>Highlight</mark>
    </p>

    <p>
      H<sub>2</sub>0
      and
      x<sup>2</sup>
    </p>

    <hr>

    <h2>Links</h2>

    <a
      href="https://github.com"
      target="_blank"
    >
      Open GitHub
    </a>

    <br><br>

    <a
      href="https://github.com"
      target="_blank"
```

```
>
  
</a>

</section>

<section id="media">

  <h2>Media</h2>

  

  <br>

  <video
    src="movie.mp4"
    width="600"
    controls
    muted
    autoplay
    loop
  >
    Your browser does not support video.
  </video>

  <audio
    src="song.mp3"
    controls
  ></audio>

</section>

<section>
```

```

<h2>Lists</h2>

<h3>Ordered</h3>

<ol>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>

<h3>Unordered</h3>

<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>

<h3>Description</h3>

<dl>
  <dt>HTML</dt>
  <dd>Structure of the web page</dd>

  <dt>CSS</dt>
  <dd>Styling of the web page</dd>
</dl>

</section>

<section>

  <h2>Table</h2>

  <table border="1">

    <thead>
      <tr>
        <th>Name</th>
        <th>Age</th>
      </tr>
    </thead>
  </table>

```

```
<tbody>
  <tr>
    <td>Ahmed</td>
    <td>25</td>
  </tr>
</tbody>

<tfoot>
  <tr>
    <td colspan="2">
      End of Table
    </td>
  </tr>
</tfoot>

</table>

</section>

<section id="form">

  <h2>Form</h2>

  <form
    action="/submit"
    method="POST"
  >

    <label for="name">
      Name:
    </label>

    <input
      type="text"
      id="name"
      name="name"
      placeholder="Your full name"
      required
    >

    <br><br>
```

```
<label for="email">
  Email:
</label>

<input
  type="email"
  id="email"
  name="email"
  placeholder="example@mail.com"
  required
>

<br><br>

<label for="phone">
  Phone:
</label>

<input
  type="tel"
  id="phone"
  name="phone"
  placeholder="+20 100 000 0000"
>

<br><br>

<label for="password">
  Password:
</label>

<input
  type="password"
  id="password"
  name="password"
>

<p>Gender:</p>

<input
  type="radio"
```

```
    id="male"
    name="gender"
    value="male"
  >

  <label for="male">
    Male
  </label>

  <input
    type="radio"
    id="female"
    name="gender"
    value="female"
  >

  <label for="female">
    Female
  </label>

  <p>Hobbies:</p>

  <input
    type="checkbox"
    id="reading"
    name="hobby"
    value="reading"
  >

  <label for="reading">
    Reading
  </label>

  <input
    type="checkbox"
    id="coding"
    name="hobby"
    value="coding"
  >

  <label for="coding">
    Coding
```

```
</label>

<br><br>

<label for="country">
  Country:
</label>

<select
  id="country"
  name="country"
>
  <option value="egypt">
    Egypt
  </option>

  <option value="usa">
    USA
  </option>
</select>

<br><br>

<label for="message">
  Message:
</label>

<textarea
  id="message"
  name="message"
  rows="5"
  cols="40"
  placeholder="Write your message..."
></textarea>

<br><br>

<button type="submit">
  Send
</button>

<input
```

```
        type="reset"
        value="Reset"
    >

    </form>

</section>

</main>

<footer>
    <p>HTML Study Notes</p>
</footer>

</body>

</html>
```

23. Quick Revision

Structure

```
<!DOCTYPE html>
<html>
<head>
<body>
```

Text

```
<h1> → <h6>
<p>
<br>
<hr>
<b>
<i>
<u>
<mark>
<del>
<sup>
<sub>
```

Links & Images

```
<a>
<img>
```

Main attributes:

```
href
target
src
alt
width
```

Layout

```
<div>  
<span>
```

Semantic

```
<header>  
<main>  
<nav>  
<section>  
<footer>
```

Media

```
<img>  
<video>  
<audio>  
<iframe>
```

Lists

```
<ul>  
<ol>  
<dl>  
<dt>  
<dd>  
<li>
```

Tables

```
<table>  
<thead>  
<tbody>  
<tfoot>  
<tr>  
<th>  
<td>
```

Important:

```
colspan → horizontal  
rowspan → vertical
```

Forms

```
<form>  
<input>  
<label>  
<select>  
<option>  
<textarea>  
<datalist>  
<button>
```

Common inputs:

```
text
password
email
file
number
range
search
tel
date
time
week
color
radio
checkbox
submit
reset
```

Important attributes:

```
placeholder
required
readonly
disabled
name
value
type
action
method
```

Final Memory Map

```
HTML
├── Structure
│   ├── DOCTYPE
│   ├── html
│   ├── head
│   └── body
├── Text
│   ├── h1-h6
│   ├── p
│   ├── br
│   ├── hr
│   └── formatting
├── Links & Images
│   ├── a
│   └── img
├── Layout
│   ├── div
│   └── span
├── Semantic
│   ├── header
│   ├── nav
│   ├── main
│   ├── section
│   └── footer
├── Media
│   ├── video
│   ├── audio
│   └── iframe
├── Lists
│   ├── ul
│   ├── ol
│   └── dl
├── Tables
│   └── table
```

```
|
|
| tr
| th
| td
|
| Forms
| input
| label
| select
| textarea
| datalist
| button
```